409.

ON THE CONDITIONS FOR THE EXISTENCE OF EQUAL ROOTS OF A BINARY QUARTIC OR QUINTIC.

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The relation for the quadric factor, for any value whatever of n, is at once seen to be expressible by means of an oblong matrix, giving rise to a series of determinants which are each to be put $= 0$; the relation for three equal roots and that for two pairs of equal roots, in the particular cases $n = 4$ and $n = 5$, are given in my "Memoir on the Conditions for the existence of given Systems of Equalities between the roots of an Equation," Phil. Trans, vol. CXLVII. (1857), pp. 727–731, [150]; and I propose in the present Memoir to exhibit, for the cases in question $n = 4$ and $n = 5$, the connexion between the compound relation for the quadric factor with the component relations for the three equal roots and for the two pairs of equal roots respectively.

Article Nos. 1 to 8. — The Quartic.

1. For the quartic function $(a, b, c, d, e \,|\, x, y)^4$, the condition for three equal roots, or, say, for a root system 31, is that the quadrinvariant and the cubinvariant each of them vanish, viz. we must have

$$I = ae - 4bd + 3c^2 = 0, \qquad J = ace - ad^2 - be^2 + 2bcd - c^3 = 0.$$

2. The condition for two pairs of equal roots, or for a root system 22, is that the cubicovariant vanishes identically, viz. representing this by

$$(A, B, 5C, 10D, 5E, F, G \,|\, x, y)^6 = 0,$$

$$\begin{aligned}
A &= a^2d - 3abc + 2b^3 &&= 0,\\
B &= a^2e + abd - 9ac^2 + 6b^2c &&= 0,\\
C &= abe - 3acd + 2b^2d &&= 0,\\
D &= -ad^2 + b^2e &&= 0,\\
E &= -ade + 3bce - 2bd^2 &&= 0,\\
F &= -ae^2 - 2bde + 9c^2e - 6cd^2 &&= 0,\\
G &= -be^2 + 3cde - 2d^3 &&= 0.
\end{aligned}$$

3. But the condition for the common quadric factor is

$$\begin{vmatrix} a, & 3b, & 3c, & d \\ b, & 3c, & 3d, & e \end{vmatrix} = 0,$$

and the determinants formed out of this matrix must therefore vanish for $(I, J) = 0$, and also for $(A, B, C, D, E, F, G) = 0$, that is, the determinants in question must be syzygetically related to the functions (I, J), and also to the functions (A, B, C, D, E, F, G).

4. The values of the determinants are

$$\begin{aligned}
1234 &= 3 \times (a^2ce + a^2d^2 - 3ab^2e + 4abcd - ac^3 - b^3d),\\
1235 &= 3 \times (a^2de - ab^2e + 2abde - 3ac^2d - ad^3 + 4b^2cd - b^3e),\\
1245 &= (a^2e^2 - abde + 2ac^2e - ad^2e - 3b^2ce + 4bc^2d - c^4),\\
1345 &= 3 \times (abce - abde + 2ac^2e - 3ad^2e - 9b^2de + 14bcd^2 - c^3d),\\
2345 &= 3 \times (ace^2 - ade^2 + 2bce^2 - 3cde^2 + 6d^2e - c^3e - d^4).
\end{aligned}$$

5. The syzygetic relation with (I, J) is given by means of the identical equation

$$(1234, 1235, 1245, 1345, 2345 \,|\, x, y)^4 = -6I \cdot HU + 9J \cdot U,$$

where HU is the Hessian of U,

$$HU = (ac - b^2, ad - bc, ae + 2bd - 3c^2, be - cd, ce - d^2 \,|\, x, y)^4,$$

or, as this may be written,

$$(1234, 1235, 1245, 1345, 2345 \,|\, x, y)^4 = -6I \cdot HU + 9J \cdot U.$$

6. That is, we have

$$\begin{aligned}
1234 &= (ac - b^2,\ a^2d - 6bI,\ 9J),\\
1235 &= (2ad - 2bc,\ 4bd - 6cI,\ 9J),\\
1245 &= (ae + 2bd - 3c^2,\ 6ce - 6dI,\ 9J),\\
1345 &= (2be - 2cd,\ 4d^2 - 6eI,\ 9J),\\
2345 &= (ce - d^2,\ e^2 - 6fI,\ 9J).
\end{aligned}$$

7. The determinants thus vanish if $(I, J) = 0$, that is, for the root system 31; they will also vanish without this being so, if only

$$\frac{3J}{2I} = \frac{ac - b^2}{a} = \frac{ad - bc}{2b} = \frac{ae + 2bd - 3c^2}{6c} = \frac{be - cd}{2d} = \frac{ce - d^2}{e},$$

and we may omit the first member $\left(\frac{3J}{2I}\right)$, since if the remaining terms are equal to each other they will also be $= \frac{3J}{2I}$. The equations may then be written

$$\frac{ac - b^2}{a} = \frac{ad - bc}{2b} = \frac{ae + 2bd - 3c^2}{6c} = \frac{be - cd}{2d} = \frac{ce - d^2}{e},$$

and the ten equations of this system reduce themselves (as it is very easy to show) to the seven equations

$$(A, B, C, D, E, F, G) = 0,$$

which, as above mentioned, are the conditions for the root system 22.

8. It may be added that we have

1234	1235	1245	1345	2345
c	c	$-e + 4d - 3c$	$-e + 3d - c$	$-3e + 4d - c$
$-4b + 3a$	$-3b + a$	$= d - 3c + a$	$= d - 3c + b$	
	$= d - 3c + a$	$= -e + 6c - a$	$= -e + 3c - b$	
			$= -d + 3c - b$	

where it is to be noticed that the four equations having the left-hand side $= 0$, give $B : C : D : E : F$ proportional to the determinants of the matrix

$$\begin{matrix}
d, & -3c, & \cdot, & a\\
-e, & \cdot, & 6c, & \cdot\\
-a, & -d, & 3c, & -b\text{'}\\
-e, & \cdot, & +3c, & -b
\end{matrix}$$

the determinants in question contain each the factor c, and omitting this factor, the system shows that B, C, D, E, F are proportional to their before-mentioned actual values.

Article Nos. 9 to 15. — The Quintic.

9. For the quintic function $(a, b, c, d, e, f \mid x, y)^5$, the condition of a root system 41 is that the covariant, $[B =]$ No. 14, shall vanish, viz. we must have

$$A = 2(ae - 4bd + 3c^2) = 0, \quad B = af - 3be + 2cd = 0, \quad C = 2(bf - 4ce + 3d^2) = 0.$$

10. The condition of a root system 32 is that the following covariant, viz.

$$[25A'B - 25G =]3(\text{No. } 13)^2(\text{No. } 14) - 25(\text{No. } 15)^2,$$

shall vanish; or, as this may be written, that we must have

$$\begin{vmatrix} A, & B, & C \\ A_1, & B_1, & C_1 \\ A_2, & B_2, & C_2 \end{vmatrix} = 0,$$

where A_1, B_1, C_1 and A_2, B_2, C_2 are the first and second differential coefficients of A, B, C respectively.

11. The condition of a root system $2^2 1$ is that the quadricovariant and the quarticovariant shall each vanish identically; or, what is the same thing, that we must have

$$(21, 33, \alpha\beta, \alpha\gamma, \ldots \mid x, y)^8 = 0.$$

12. The conditions for the common [cubic] factor are

$$\begin{vmatrix} a, & 4b, & 6c, & 4d, & e & \\ & a, & 4b, & 6c, & 4d, & e \\ b, & 4c, & 6d, & 4e, & f & \\ & b, & 4c, & 6d, & 4e, & f \end{vmatrix} = 0,$$

the several determinants whereof are given in Table No. 27 of my "Third Memoir on Quantics," Philosophical Transactions, vol. CXLVI. (1856), pp. 627–647, [144].

13. These determinants must therefore vanish, for $(A, B, C) = 0$, and also for $(21, 33, \ldots 8, 9) = 0$, that is, they must be syzygetically connected with (A, B, C), and also with $(21, 33, \ldots 8, 9)$. The relation to (A, B, C) is in fact given in the Table appended to Table No. 27, viz. this is

$C\times$	$B\times$	$A\times$	
$1234 = +$	$+6a^2 +$	$-12ab +$	$+16ac - 10b^2$
$1235 = +$	$+6ab +$	$-2ac - 10b^2 +$	$+6bc$
$1236 = +$	$-2ac + 8b^2 +$	$+6ad - 18bc +$	$-2ae + 8bd$
$1245 = +$	$+18ac +$	$-6ad - 30bc +$	$+8ae + 10bd$
$1246 = +$	$+12bc +$	$+4ae - 4bd - 24c^2 +$	$+4be + 8cd$
$1345 = +$	$+24ad +$	$-8ae - 40bd +$	$+4af + 20be$
$1256 = +$	$-1ae + 4bd + 3c^2 +$	$+af + 5be - 18cd +$	$-16bf + 4ce + 3df$
$2345 = +$	$+20ae + 40bd - 30c^2 +$	$-80be + 20cd + 20bf + 40ce - 30d^2 +$	
$1346 = +$	$+4ae + 8bd + 6c^2 +$	$-36cd + bf + 8ce + 9d^2 +$	
$2346 = +$	$+4af + 20be +$	$-8bf - ace + 24c^2 +$	
$1356 = +$	$+4be + 8cd +$	$+4bf - 4ce - 24d^2 + 12de +$	
$2356 = +$	$+8bf + 10ce +$	$-6cf - 30de + 18d^2 +$	
$1456 = +$	$+6ce + 6cf - 18de +$	$-2df + 8e^2 +$	
$2456 = +$	$+6cf - 2df - 10e^2 +$	$+6ef +$	
$3456 = +$	$+16df - 10e^2 +$	$-12ef + 6f^2 +$	

14. Between the expressions $21, 33$, &c., and $1234, 1235$, &c., there exist relations the form of which is indicated by the following Table:

15. Assuming the existence of these relations, we have for the determination of the numerical coefficients in each relation a set of linear equations, which are shown by the following Tables, viz. referring to the Table headed $c21$, $d33$, $a\beta$, $a.1234$, [first of the seven tables infra] if the multipliers of the several terms respectively be A, B, C, X, then the Table denotes the system of linear equations

$$A + 3B + 33C + 0X = 0, \quad 3A + 0B - 10C - 16X = 0, \quad \&\text{c.},$$

that is, nine equations to be satisfied by the ratios of the coefficients A, B, C, X, and which are in fact satisfied by the values at the foot of the Table, viz. $A : B : C : X = +66 : -11 : +1 : +6$.

There would be in all fourteen Tables, but as those for the second seven would be at once deducible by symmetry from the first seven, I have only written down the seven Tables; the solutions for the first and second Tables were obtained without difficulty, but that for the third Table was so laborious to calculate, and contains such extraordinarily high numbers, that I did not proceed with the calculation, and it is accordingly only the first, second, and third Tables which have at the foot of them respectively the solutions of the linear equations.

16. The results given by these three Tables are, of course,

$$66c21 - 11d33 + 1a\beta + 6a.1234 = 0,$$
$$330c21 + 110d33 - 55a\beta + 9a\gamma - 105a.1235 = 0,$$
$$266478575c21 - 617359490d33 + 144200810e\beta + 9656911e\gamma + 9090785a\delta - 7210040a\epsilon = 0.$$

410.

A THIRD MEMOIR ON SKEW SURFACES, OTHERWISE SCROLLS.

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The present Memoir is supplementary to my "Second Memoir on Skew Surfaces, otherwise Scrolls," Phil. Trans, vol. CLIV. (1864), pp. 559–577, [340], and relates also to the theory of skew surfaces of the fourth order, or quartic scrolls.

It was pointed out to me by Herr Schwarz[1], in a letter dated Halle, June 1, 1867, that in the enumeration contained in my Second Memoir I have given only a particular case of the quartic scrolls which have a directrix skew cubic; viz. my eighth species, $S(1, 3^2)$, where there is also a directrix line. And this led me to observe that I had in like manner mentioned only a particular case of the quartic scrolls with a triple directrix line; viz. my third species, $S(1^3, 1, 4)$, where there is also a simple directrix line. The omitted species, say, ninth species, $S(1^3)$, with a triple directrix line, and tenth species, $S(3^2)$, with a directrix skew cubic, are considered in the present Memoir; and in reference to them I develope a theory of the reciprocal relations of these scrolls, which has some very interesting analytical features.

The paragraphs of the present Memoir are numbered consecutively with those of my Second Memoir above referred to.

Quartic Scroll, Ninth Species, $S(1^3)$, with a triple directrix line.

54. Consider a line the intersection of two planes, and let the equation of the one plane contain in the order 3, that of the second plane contain linearly, a variable parameter θ; the equations of the two planes may be taken to be

$$(p, q, r, s \,|\, \theta, 1)^3 = 0, \qquad (u, v \,|\, \theta, 1) = 0,$$

where (p, q, r, s, u, v) are any linear functions whatever of the coordinates (x, y, z, w). Hence eliminating θ we have as the equation of the scroll generated by the line in question $(p, q, r, s \,|\, v, -u)^3 = 0$, viz. this is a quartic scroll having the line $u = 0$, $v = 0$ for a triple line; that is, the line in question is a triple directrix line.

55. Taking $x = 0$, $y = 0$ for the equations of the directrix line, or writing $u = x$, $v = y$, and moreover expressing (p, q, r, s) as linear functions of the coordinates (x, y, z, w), the equation of the scroll takes the form

$$(* \,|\, x, y)^3 z + (* \,|\, x, y)^2 w + (* \,|\, x, y) z^2 + (* \,|\, x, y) zw + k(x, y)^4 = 0;$$

and we may, by changing the values of z and w, make the term in $(x, y)^4$ to be

$$(* \,|\, x, y)^4 + (\alpha z + \beta w)(* \,|\, x, y)^3 + (\gamma z + \delta w)(* \,|\, x, y)^2,$$

where the arbitrary constants $\alpha, \beta, \gamma, \delta$ may be so determined as to reduce this to a monomial kx^4, kx^3y, or kxy^3.

56. The coefficient k may vanish, and the equation of the scroll then is

$$(* \,|\, x, y)^3 z + (* \,|\, x, y)^2 w = 0, \quad \text{or, what is the same thing,} \quad (* \,|\, x, y)^2 (z, w) = 0,$$

viz. the scroll has in this particular case the simple directrix line $z = 0$, $w = 0$, thus reducing itself to the third species, $S(1^3, 1, 4)$, with a triple directrix line and a single directrix line. It is proper to exclude this, and consider the ninth species, $S(1^3)$, as having a triple directrix line, but no simple directrix line.

57. The scroll $S(1^3)$ may be considered as a scroll $S(m, n, p)$ generated by a line which meets each of three given directrices; viz. these may be taken to be the directrix line, and any two plane sections

[1] I take the opportunity of referring to his paper on Quintic Scrolls, Schwarz, "Ueber die geradlinigen Flächen fünften Grades," Crelle, t. LXVII. (1867), pp. 23–57.

of the scroll. The section by any plane is a quartic curve having a triple point at the intersection with the directrix line; moreover the sections by any two planes meet in four points, the intersections of the scroll by the line of intersection of the two planes. Conversely, taking any line and two quartics related as above (that is, each quartic has a triple point at its intersection with the directrix line, and the two quartics meet in four points), the scroll generated by the line meeting these three directrices is a quartic scroll having the line as a triple directrix line; and it is moreover such that through each point of the line there pass three generating lines, but through each point of either of the plane quartics only a single generating line; that is, that the line is a triple directrix line, but each of the plane quartics a simple directrix curve.

58. We may instead of the section by any plane, consider the section by a plane through a generating line, or by a plane through two of the three generating lines which meet at any point of the directrix line; if (to consider only the most simple case) each of the planes be thus a plane through two generating lines, the section by either of these planes is made up of the two generating lines, and of a conic passing through the directrix line; the directrices are thus the line and two conics each of them meeting the line; we have therefore in the foregoing formula $m = 1$, $n = 2$, $p = 2$, $\alpha = 0$, $\beta = 1$, $\gamma = 1$, and the order of the scroll is $8 - 2 - 2 = 4$ as before.

Quartic Scroll, Tenth Species, $S(3^2)$, with a directrix skew cubic met twice by each generating line.

59. Consider a line, the intersection of two planes; and let the equation of each plane contain in the order 2 a variable parameter θ; the equations of the two planes may be taken to be

$$(p, q, r \,|\, \theta, 1)^2 = 0, \qquad (p', q', r' \,|\, \theta, 1)^2 = 0,$$

where (p, q, r, p', q', r') are linear functions of the coordinates (x, y, z, w); hence eliminating θ, we have as the equation of the scroll generated by the line in question, $D = 0$, where D is the resultant of the two quadric functions.

The equation may be written

$$4(pq' - p'q)(rq' - r'q) - (pr' - p'r)^2 = 0;$$

and the scroll has thus as a nodal (double) line the skew cubic determined by the equations

$$\begin{vmatrix} p, & q, & r \\ p', & q', & r' \end{vmatrix} = 0.$$

It is easy to see (and indeed it will be shown presently) that this curve is met twice by each generating line of the scroll, and that the scroll is consequently a quartic scroll as described above.

60. The coordinates (x, y, z, w) may be fixed in such manner that the equations of the skew cubic shall be

$$xw - yz = 0, \quad xz - y^2 = 0, \quad yw - z^2 = 0;$$

or, what is the same thing,

$$yw - z^2 = 0, \quad zy - xw = 0, \quad xz - y^2 = 0;$$

each of the equations $pq' - p'q = 0$, $rq' - r'q = 0$, $pr' - p'r = 0$ is then the equation of a quadric surface passing through the skew cubic, or, what is the same thing, each of the functions $pq' - p'q$, $rq' - r'q$, $pr' - p'r$ is a linear function of $yw - z^2$, $zy - xw$, $xz - y^2$; and the equation of the scroll is given as a quadric equation in the last-mentioned quantities.

It will be convenient to represent the equation in the form

$$(H, F, G, B, A - F, -G \,|\, yw - z^2, zy - xw, xz - y^2)^2 = 0,$$

or, writing for shortness $yw - z^2$, $zy - xw$, $xz - y^2 = p, q, r$, which letters (p, q, r) are used henceforward in this signification only, the equation will be

$$(H, F, G, B, A - F, -G \,|\, p, q, r)^2 = 0,$$

viz. this is a quadric equation in (p, q, r), with arbitrary coefficients.

61. Comparing with the result, Second Memoir, Nos. 47 to 50, we see that in the particular case where the coefficients (A, B, C, F, G, H) satisfy the relation

$$AF + BG + CH = 0,$$

we have the eighth species, $S(1, 3^2)$, with a directrix line and a directrix skew cubic met twice by each generating line. We exclude this particular case, and in the tenth species consider the relation $AF + BG + CH = 0$ as not satisfied, and therefore the scroll as not having a directrix line.

62. I consider how the scroll may be obtained as a scroll $S(m^n, n)$ generated by a line meeting a curve of the order m in n times and a curve of the order n once. The first curve will be the skew cubic, that is $m = 3$; the second curve may be any plane section of the scroll; such a section will be a quartic curve having three nodes, one at each intersection of its plane with the skew cubic. Conversely, if we have a skew cubic, and a plane quartic meeting the skew cubic in three points, each of them a node on the quartic, then the scroll generated by the line meeting the skew cubic twice and the plane quartic once, is a quartic scroll $S(3^2)$ as described above.

63. We may, instead of the section by a plane in general, consider the section by a plane through a generating line; the section is here made up of the generating line and of a plane cubic passing through each of the two points of intersection of the generating line with the skew cubic, and having a node at the remaining intersection of its plane with the skew cubic. Or we may consider the section by a plane through the two generating lines at any point of the skew cubic; the section is here made up of the two generating lines and of a conic passing through the second intersections of the two generating lines with the skew cubic; that is, meeting the skew cubic twice.

64. Conversely, consider a skew cubic, and a conic meeting it twice; the lines which meet the skew cubic twice, and also the conic, generate a quartic scroll; this appears by the before-mentioned formula $S(m^n, n) = n([m]^n + \mu) -$ reduction; viz. we have $m = 3$, $n = 2$, and the order is $= 8 -$ reduction; the reduction arises from the cones having their vertices at the intersections of the skew cubic and the conic. Each cone is of the order 2, and (quâ simple point on the conic) each intersection gives a reduction $=$ order of the cone; that is, the total reduction is $= 4$, and the order of the scroll is $8 - 4 = 4$ as above.

65. But a more elegant mode of generation of the scroll may be obtained by means of the skew cubic alone; viz. considering the system of lines which are in involution with five given lines, or say simply the lines which belong to an involution[2], I say that the locus of a line belonging to the involution, and meeting the skew cubic twice is the quartic scroll, tenth species, $S(3^2)$.

In the particular case where the line (instead of belonging to a proper involution) passes through a fixed point (that is, where the involution is made up of the lines through the fixed point), the locus is the quadric surface through the skew cubic; that is, we have in this particular case not a quartic scroll, but a quadric surface.

66. The analytical theory presents some interesting features. I write

$$\begin{aligned} U &= (H, F, G, B, A - F, -G \,|\, p, q, r)^2 \\ &= Hp^2 + 2Fpq + 2Gpr + Bq^2 + 2(A - F)qr - Gr^2. \end{aligned}$$

The equation of the reciprocal scroll is obtained by means of the coefficients (A, B, C, F, G, H); and I write these in the form

$$\begin{aligned} A &= a, \quad B = b, \quad C = c, \\ F &= f, \quad G = g, \quad H = h. \end{aligned}$$

The reciprocal relation is obtained by expressing that the coefficients (a, b, c, f, g, h) satisfy the same equation as the coefficients (A, B, C, F, G, H); that is, that the two sets of coefficients are related in the same way as the original and reciprocal surfaces.

67. I do not pursue the analytical theory further, but content myself with having established the existence of the two new species of quartic scrolls, and having indicated their reciprocal relations. The theory of the reciprocal relations of these scrolls presents some very interesting analytical features, which I hope to develope on a future occasion.

[2] I have worded this heading in accordance with that of the eighth species, Second Memoir, No. 51.

411.

A MEMOIR ON THE THEORY OF RECIPROCAL SURFACES.

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1. The present Memoir is supplementary to my "Memoir on the Theory of Reciprocal Surfaces," Phil. Trans, vol. CLVII. (1867), pp. 201–229, [380]. I take account of conical and biplanar nodes, or, as I call them, cnicnodes, and binodes; of pinch-points on the nodal curve; and of close-points and off-points on the cuspidal curve: viz. I assume that there are

C, cnicnodes,
B, binodes,
j, pinch-points,
χ, close-points,
θ, off-points,

deferring for the present the explanation of these singularities. The same letters, accented, refer to the reciprocal singularities. Or using "trope" as the reciprocal term to node, these will be C', cnictropes, B', bitropes, j', pinch-planes, χ', close-planes, θ', off-planes; but these present themselves, not in the equations above referred to, but in the reciprocal equations.

2. I take account of conical and biplanar nodes, or, as I call them, cnicnodes, and binodes; of pinch-points[3] on the nodal curve; and of close-points and off-points on the cuspidal curve: viz. I assume that there are C, cnicnodes, B, binodes, j, pinch-points, χ, close-points, θ, off-points, deferring for the present the explanation of these singularities.

3. The resulting alterations are that we must in the formulae write $\kappa - B$, $B - C$ in place of κ, B respectively; and change the formulae for $c(n-2)$, $[ab]$, $[bc]$, into

$$\begin{aligned} c(n-2) &= 2\sigma + \chi + \theta + \delta, \\ [ab] &= ab - 2p - j, \\ [ac] &= ac - 3\sigma - \chi. \end{aligned}$$

4. Making these changes, and substituting for $[ab]$, $[ac]$, $[bc]$ their values, the formulae become

$$\begin{aligned} a(n-2) &= \kappa - B + p + 2\sigma, \\ b(n-2) &= p + 2\rho + 3\chi + 3\theta, \\ c(n-2) &= 2\sigma + \chi + \theta + \delta, \\ a(n-2)(n-3) &= 2(\delta - \theta) + 3(ac - 3\sigma - \chi) + 2(ab - 2p - j), \\ b(n-2)(n-3) &= \kappa + (ab - 2p - j) + 3(bc - 3\rho - 2\chi - i), \\ c(n-2)(n-3) &= 6h + (ac - 3\sigma - \chi) + 2(bc - 3\rho - 2\chi - i), \end{aligned}$$

which replace the original formulae (A) and (B).

5. For convenience I annex the remaining equations; viz. these are

$$\begin{aligned} a' &= n(n-1) - 2b - 3c, \\ \kappa' &= 3n(n-2) - 6b - 8c, \\ \delta' &= \tfrac{1}{2}n(n-2)(n^2-9) - (n^2-n-6)(2b+3c) + 2b(b-1) + 6bc + \tfrac{3}{2}c(c-1); \end{aligned}$$

the equations

$$q = \tfrac{1}{2}a - b - 2\kappa - 3j - 6\theta, \qquad r = c - c - 2h - 3\rho,$$

[3] This addition to the theory is in fact indicated in Salmon, see the note, p. 445; the i there employed, which

(q, r in place of Salmon's R, S respectively); the equation $\sigma = \sigma$; and the corresponding equations, interchanging the accented and unaccented letters, in all 23 equations between the 42 quantities

$$n, a, B, \kappa; \quad b, k, t, q, p, j; \quad c, h, r, \sigma, \rho, \chi, \theta, i;$$
$$B, C; \quad n', a', \kappa', \delta'; \quad b', k', t', q', p', j'; \quad c', h', r', \sigma', \rho', \chi', \theta', i'; \quad B', C'.$$

6. We have

$$(a - b - c)(n - 2) = (\kappa - B - \delta) - 6\rho - \chi - 3\theta,$$
$$(a - 2b - 3c)(n - 2)(n - 3) = 2(\delta - \theta) - 8\kappa - 18h - 6bc - 8\rho - 2\chi - i);$$

and substituting these values of δ, κ in the formula $n' = a(a - 1) - 2\delta - 3\kappa$, and for a its value, $= n(n - 1) - 2b - 3c$, we find

$$n' = n(n - 1)^2 - n(7b + 12c) + 4b^2 + 8b + 9c^2 + 15c$$
$$- 8\kappa - 18h + 18\rho + 12\chi + 12i - 9\theta$$
$$- 2C - 3B - 3\Theta,$$

where the foregoing equations for $a - b - c$ and $a - 2b - 3c$ show clearly the origin of the new terms $-2C - 3B - 3\Theta$; these express that there is in the value of n' a reduction $= 2$ for each cnicnode, $= 3$ for each binode, and $= 3$ for each off-point.

7. We have $(n - 2)(n - 3) = n^2 - n + (-4n + 6) = a + 2b + 3c + (-4n + 6)$; and making this substitution in the equations which contain $(n - 2)(n - 3)$, these become

$$a(-4n + 6) = 2(\delta - \theta) - a^2 - 4p - 9\sigma - 2j - 3\chi,$$
$$b(-4n + 6) = 4\kappa - 2b^2 - 9\rho - 6\chi - 8i + 2p - j,$$
$$c(-4n + 6) = 6h - 3c^2 - 6\rho - \chi - 2i - 2\theta,$$

(Salmon's equations (C)); and adding to each equation 4 times the corresponding equation with the factor $(n - 2)$, these become

$$a^2 - 2a = 2(\delta - \theta) + (\kappa - B) - a^2 - 2j - 3\chi,$$
$$2b^2 - 2b = 4\kappa - \rho + 6\chi + 12\theta - 3i + 2p - j,$$
$$3c^2 - 2c = 6h + 10\rho + 4\chi - 2i + 5\sigma - \chi.$$

Writing in the first of these $a^2 - 2a = a(a - 1) - a, = n' + 2\delta + 3\kappa - a$, and reducing the other two by means of the values of q, r, the equations become

$$n' - a = -2C - 4B + \kappa - a - 2j - 3\chi,$$
$$2q + \chi + 3i + j = 2p,$$
$$3r + c + 2i + \chi = 5\sigma + \rho + 4\theta,$$

(Salmon's equations (D)).

I attend in particular to the first of these, or rather to the reciprocal equation, which will be

$$\sigma' = a - n + \kappa' - 2j' - 3\chi' - 2C' - 4B',$$

which, writing therein $a = n(n - 1) - 2b - 3c$, and $\kappa = 3n(n - 2) - 6b - 8c$, becomes

$$\sigma' = 4n(n - 2) - 8b - 11c - 2j' - 3\chi' - 2C' - 4B'.$$

The singularity σ' is not explicitly defined in Salmon; σ' is the reciprocal of σ, and (as such) it denotes the number of common tangent planes of the spinode torse and of the torse generated by the tangent planes along a plane section of the surface; or, what is the same thing, it is the number of the spinode planes which touch the plane section; that is, it is equal to the number of points of intersection of the spinode curve and the plane section; or, finally σ' is the order of the spinode curve.

The spinode curve is in fact for a surface of the order n without singularities the intersection of the surface by the Hessian surface of the order $4(n-2)$, and is thus a curve of the order $4n(n-2)$; or, introducing the singularities b, c, it is a curve of the order $4n(n-2)-8b-11c$.

8. It is clear that p will in like manner denote the order of the node-couple curve.

9. I express in terms of n, b, c, h, k, ρ, χ, j, θ, χ, C, B such quantities and combinations of quantities as can be so expressed. We have

$$\begin{aligned}
a = a' &= n(n-1)-2b-3c,\\
\kappa &= 3n(n-2)-6b-8c,\\
\delta' &= \tfrac{1}{2}n(n-2)(n^2-9)-(n^2-n-6)(2b+3c)+2b(b-1)+6bc+\tfrac{3}{2}c(c-1),\\
\mu &= 12k+c(6n-6)-6c^2-5\chi+3\theta-2\rho,\\
24t &= (-8n+8)b+(10n-18)c+8b^2-18c^2-2(8\kappa-18h)+20\rho-10\chi+4j+9\chi+6\theta,\\
\delta &= b^2-b-2i-3\chi-6\theta, \quad (t \text{ supra}),\\
r &= c^2-c-2h-3\rho,\\
2\sigma &= c(n-2)-(4\rho+\chi)-\theta,\\
8p &= (16n-24)b+(-15n+18)c-8b^2+18c^2+2(8\kappa-18h)-9(4\rho+\chi)-4j-9\chi-6\theta,\\
8\kappa' &= 8n(n-1)(n-2)+b(-32n+56)+c(-42n+46)+8b^2-18c^2\\
&\quad -2(8\kappa-18h)+17(4\rho+\chi)+4j+17\chi+6\theta+8\sigma,\\
24h &= n(n-1)(n-2)(n-3)+b(-4n^2+20n-24)+c(-6n^2+18n-18)+12bc+18c^2\\
&\quad +(8\kappa-18h)-9(4\rho+\chi)-9\chi+2(7,\\
8a' &= 8n(n-1)^2+b(-32n+40)+c(-21n+30)+8b^2-18c^2\\
&\quad -2(8\kappa-18h)+21(4\rho+\chi)-12j+21\chi-18\theta-16C-24B,\\
8c' &= 4n(n-1)(n-2)+b(-16n+28)+c(-10n+26)+4b^2-9c^2\\
&\quad -(8\kappa-18h)+10(4\rho+\chi)-4j+10\chi-6\theta-6C-8B,\\
2b' &= -a+n'(n-1)-3c', \quad (n', c' \text{ supra}),\\
\sigma'+2j'+3\chi'+2C'+4B' &= 4n(n-2)-8b-11c,\\
p'-4j'-6\chi'-4C'-9B' &= -11n(n-2)+a(n'-2)+22b+30c,\\
&\quad (n', a \text{ supra}),\\
2\sigma'+4\rho'+\chi'+\theta' &= c'(n'-2), \quad (n', c' \text{ supra}),\\
4\rho'-3(2i'+3\chi'+2\theta')-2p'-j' &= (-4n'+6)c' \text{ (formulae E)}.
\end{aligned}$$

10. I express in terms of n, b, c, h, k, ρ, χ, j, θ, χ, C, B such quantities and combinations of quantities as can be so expressed.

11. Forming the combinations $4q+6r$, $24t-8q+18r$ (the last of which introduces on the opposite side the term $+48t$), we obtain

$$\begin{aligned}
4q+6r &= c(6n-12)-\sigma\chi-18\rho+3\delta-2\chi,\\
24t-8q+18r &= -(8n-16)b+(15n-36)c-34\rho+8\chi+4j+9\chi+6\theta,
\end{aligned}$$

equations which are used post, No. 53.

12. I remark that if there be on a surface a right line which is such that the tangent plane is different at different points of the line, the line is said to be *scrolar*: the section of the surface by any plane through the line contains the line once. But if there is at each point of the line one and the same tangent plane, then the section of the surface by the tangent plane contains the line at least twice; if it contain it twice only, the line is *torsal*; if three times the line is *oscular*; and the tangent plane containing the torsal or oscular line may in like manner be termed a *torsal* or an *oscular* tangent plane. These epithets, scrolar, torsal and oscular, will be convenient in the sequel.

Article Nos. 13 to 39. — Explanation of the New Singularities.

I proceed to the explanation of the new singularities.

13. The *cnicnode*, or singularity $C = 1$, is an ordinary conical point; instead of the tangent plane we have a proper quadricone.

14. The *cnictrope*, or reciprocal singularity $C' = 1$, is also a well known one; it is in fact the conic of plane contact, or say rather the plane of conic contact, viz. the cnictrope is a plane touching a surface, not at a single point, but along a conic; the conic is the intersection of the plane with a proper quadricone, the enveloping quadricone of the surface at any point of the conic.

15. For a surface having the cnicnode $C = 1$, the tangent cone is a proper quadricone; and the section of the surface by a plane through the cnicnode is a curve having at the cnicnode a node, the two tangents whereof are the intersections of the plane with the quadricone; that is, the two tangents are any two lines through the cnicnode which lie in a plane through the axis of the quadricone. But the quadricone may be such that for a certain plane or planes through the axis, the section has at the cnicnode a cusp; the plane in question is the "cuspidal plane" of the cnicnode. I do not at present attend to this singularity.

16. For a surface having the cnictrope $C' = 1$, the section by the plane of the cnictrope is made up of the conic of contact twice, and of a residual curve of the order $n - 4$; that is, the conic of contact counts as two coincident lines in the section by the plane of the cnictrope. The cnictrope is thus a particular case of a trope; and the singularities of the surface are modified accordingly.

17. For a surface having the cnictrope $C' = 1$, the Hessian surface passes through the conic, which is thus thrown off from the spinode curve; or there is a reduction $= 2$ in the order of the curve, which agrees with a foregoing result.

18. The *binode*, or singularity $B = 1$, is a biplanar node, where instead of the proper quadricone we have two planes; these may be called the *biplanes*, and their line of intersection, the *edge* of the binode. The biplanes form a plane-pair.

19. The *bitrope*, or reciprocal singularity $B' = 1$, is the plane of point-pair contact; but this needs explanation.

20. Consider a surface having a binode, and the reciprocal surface having a bitrope. We have the bitrope, a plane the reciprocal of the binode; in this plane a line, the reciprocal of the edge; in the line two points, or say a point-pair, the reciprocal of the biplanes: these points may be called the *bipoints*. There are in each biplane three directions of closest contact; the reciprocals of these are in the bitrope three directions through each of the two points. The section of the reciprocal surface by the bitrope is made up of the line counting three times (or the line is oscular), and of a curve passing in the three directions (having therefore a triple point) through each of the two bipoints. The bitrope contains thus an oscular line; but it is part of the notion that there are on this line two points each a triple point on the residual curve.

21. The section of the surface having a binode, by a plane through the edge, is made up of the edge twice (representing the two biplanes) and of a residual curve of the order $n-2$, having at the intersection with the edge a node, the two tangents whereof are the intersections of the plane with the biplanes. But if the plane coincide with either of the biplanes, then the section is made up of this biplane, and of the other biplane counting once, and of a residual curve of the order $n - 2$; that is, the section has at each point of the edge a node, the two tangents being the same for all points of the edge; or, what is the same thing, the edge is a torsal line on the surface.

22. The *pinch-point*, or singularity $j = 1$, is a point on the nodal curve at which the two tangent planes coincide; that is, it is a cuspidal point on the nodal curve. The nodal curve is in general a curve along which the surface has two distinct tangent planes; but at a pinch-point these two tangent planes coincide, and the nodal curve has at the pinch-point a cuspidal singularity.

23. The section of the surface by a plane through the tangent at a pinch-point is made up of the tangent twice (representing the two coincident tangent planes) and of a residual curve of the order $n-2$, which has at the pinch-point a cusp, the tangent at the cusp being the tangent to the nodal curve. In the particular case where the nodal curve is a right line, the section is the line twice (representing the cuspidal branch), and a residual curve of the order $n - 2$, the tangent to which is the cotangent.

24. The *pinch-plane*, or reciprocal singularity $j' = 1$, is in fact a torsal plane touching the surface along a line, or meeting it in the line twice and in a residual curve. Let the line and curve meet in a point P; for the reason that the section by the plane is the line twice and the residual curve, the section has at P two coincident nodes; that is, the plane is a node-couple plane with two coincident nodes. The plane meets the cuspidal curve in a point Q; and at this point the plane touches the cuspidal torse; that is, the plane is a torsal plane of the cuspidal torse. The point Q is the reciprocal of the pinch-point; and the line PQ is the reciprocal of the tangent at the pinch-point.

25. The *close-point*, or singularity $\chi = 1$, is a point on the cuspidal curve at which the cuspidal torse has a singularity; viz. it is a point at which the tangent plane to the cuspidal torse coincides with the tangent plane to the surface. The cuspidal curve is in general a curve along which the surface has a single tangent plane (counting twice); but at a close-point the cuspidal torse has a higher singularity, and the tangent plane to the cuspidal torse coincides with the tangent plane to the surface.

26. The section of the surface by a plane through the tangent at a close-point is made up of the tangent twice (representing the cuspidal branch) and of a residual curve of the order $n-2$, which has at the close-point a node, the two tangents at the node being coincident; that is, the residual curve has at the close-point a cusp, the tangent at the cusp being the tangent to the cuspidal curve. The close-point is thus a point at which the cuspidal curve has a higher singularity than usual.

27. The *close-plane*, or reciprocal singularity $\chi' = 1$, is a plane which touches the surface along a line, or meets it in the line twice and in a residual curve; but it is a particular case of the pinch-plane, where the two coincident nodes on the residual curve are themselves coincident; that is, where the residual curve has at the point of intersection with the line a cusp instead of a node. The close-plane is thus a particular case of a torsal plane; and the singularities of the surface are modified accordingly.

28. The *off-point*, or singularity $\theta = 1$, is a point on the cuspidal curve at which the cuspidal torse has a still higher singularity; viz. it is a point at which the cuspidal torse has a cuspidal edge passing through the point. The off-point is thus a point at which the cuspidal curve has a cuspidal singularity; and the tangent plane to the cuspidal torse at the off-point is indeterminate.

29. The section of the surface by a plane through the tangent at an off-point is made up of the tangent three times (representing the cuspidal branch with a higher singularity) and of a residual curve of the order $n-3$, which has at the off-point a triple point. The off-point is thus a point at which the cuspidal curve has a very high singularity; and the singularities of the surface are modified accordingly.

30. The *off-plane*, or reciprocal singularity $\theta' = 1$, is a plane which touches the surface along a line, or meets it in the line three times and in a residual curve; that is, it is a plane which is oscular to the surface along a line. The off-plane is thus a particular case of a torsal plane; and the singularities of the surface are modified accordingly.

31. The foregoing explanations of the new singularities are sufficient for the purposes of the present Memoir. I have given a more complete account of these singularities in my paper "On the Singularities of Surfaces," Proc. Lond. Math. Soc. vol. III. (1869), pp. 166–174, [395]; and I refer to that paper for further details.

32. The singularities C, B, j, χ, θ are connected with the singularities C', B', j', χ', θ' by the reciprocal relations which have been explained. The number of each kind of singularity is a characteristic of the surface; and the reciprocal singularities are determined by the original singularities, and vice versa.

33. The theory of the new singularities may be applied to the determination of the singularities of a surface which is given by its equation. For instance, if the surface has a cnicnode, or a binode, or a pinch-point, or a close-point, or an off-point, then the singularities of the reciprocal surface may be determined by means of the formulae which have been given.

34. The theory may also be applied to the determination of the singularities of a surface which is the envelope of a system of planes. For instance, if the surface is the envelope of a system of planes which depend on two parameters, then the singularities of the surface may be determined by means of the theory of the new singularities.

35. I give some examples of the application of the theory. First, if the surface has a cnicnode $C = 1$, then the reciprocal surface has a cnictrope $C' = 1$; and the order of the spinode curve is reduced by 2. Second, if the surface has a binode $B = 1$, then the reciprocal surface has a bitrope $B' = 1$; and the order of the node-couple curve is reduced by 3. Third, if the surface has a pinch-point $j = 1$, then the reciprocal surface has a pinch-plane $j' = 1$; and the order of the cuspidal curve is reduced by 1.

36. The theory of the new singularities is also connected with the theory of curves. For instance, the cnicnode is a particular case of a node; and the cnictrope is a particular case of a trope. The binode is a particular case of a node; and the bitrope is a particular case of a trope. The pinch-point is a particular case of a cusp; and the pinch-plane is a particular case of a torsal plane. The close-point is a particular case of a cusp; and the close-plane is a particular case of a torsal plane. The off-point is a particular case of a cusp; and the off-plane is a particular case of a torsal plane.

37. The theory of the new singularities may be extended to higher dimensions. For instance, in four-dimensional space, the cnicnode is a particular case of a node; and the cnictrope is a particular case of a trope. The binode is a particular case of a node; and the bitrope is a particular case of a trope. The pinch-point is a particular case of a cusp; and the pinch-plane is a particular case of a torsal plane. The close-point is a particular case of a cusp; and the close-plane is a particular case of a torsal plane. The off-point is a particular case of a cusp; and the off-plane is a particular case of a torsal plane.

38. The theory of the new singularities is a very beautiful and important branch of geometry; and it has many applications to the theory of curves and surfaces. The principles which I have explained are sufficient to determine the singularities of any surface and its reciprocal; and they may be applied to a great variety of problems.

39. I have now given a sufficient account of the theory of the new singularities; and I conclude this part of the Memoir with a few remarks on the connection of the theory with other branches of geometry. The theory of the new singularities is connected with the theory of curves, with the theory of congruences, with the theory of complexes, and with the theory of transformations. It is also connected with the theory of invariants and covariants; and it has many applications to the physical sciences.

Article Nos. 40 to 53. — Further Developments.

40. I proceed to further developments of the theory of reciprocal surfaces. The formulae which have been given may be used to determine the singularities of a surface which is given by its equation; and they may also be used to determine the singularities of a surface which is the envelope of a system of planes.

41. The formulae for the singularities of a surface of the order n with singularities b, c, h, k, ρ, χ, j, θ, C, B are as follows:

$$\begin{aligned}
&a = n(n-1) - 2b - 3c,\\
&\kappa = 3n(n-2) - 6b - 8c,\\
&\delta' = \tfrac{1}{2}n(n-2)(n^2-9) - (n^2-n-6)(2b+3c) + 2b(b-1) + 6bc + \tfrac{3}{2}c(c-1),\\
&\mu = 12k + c(6n-6) - 6c^2 - 5\chi + 3\theta - 2\rho.
\end{aligned}$$

42. The formulae for the reciprocal singularities are:

$$\begin{aligned}
&a' = n'(n'-1) - 2b' - 3c',\\
&\kappa' = 3n'(n'-2) - 6b' - 8c',\\
&\delta = \tfrac{1}{2}n'(n'-2)(n'^2-9) - (n'^2-n'-6)(2b'+3c') + 2b'(b'-1) + 6b'c' + \tfrac{3}{2}c'(c'-1).
\end{aligned}$$

43. The formulae for the order of the spinode curve and the node-couple curve are:

$$\begin{aligned}
\sigma' &= 4n(n-2) - 8b - 11c - 2j' - 3\chi' - 2C' - 4B',\\
p &= (16n-24)b + (-15n+18)c - 8b^2 + 18c^2 + 2(8\kappa - 18h)\\
&\quad - 9(4\rho+\chi) - 4j - 9\chi - 6\theta.
\end{aligned}$$

44. The formulae for the order of the cuspidal curve and the cuspidal torse are:

$$\begin{aligned}
c' &= 4n(n-1)(n-2) + b(-16n+28) + c(-10n+26) + 4b^2 - 9c^2\\
&\quad - (8\kappa - 18h) + 10(4\rho+\chi) - 4j + 10\chi - 6\theta - 6C - 8B,\\
r &= c^2 - c - 2h - 3\rho.
\end{aligned}$$

45. The formulae which have been given are sufficient to determine the singularities of any surface and its reciprocal. They may be used to verify the results which have been obtained; and they may also be used to determine the singularities of surfaces which have not been previously considered.

46. The theory of reciprocal surfaces is a very beautiful and important branch of geometry. It has many applications to the theory of curves and surfaces; and it is also connected with the theory of invariants and covariants. The principles which I have explained are sufficient to determine the singularities of any surface and its reciprocal.

47. I have now given a sufficient account of the theory of reciprocal surfaces; and I conclude this Memoir with a few remarks on the connection of the theory with other branches of geometry. The theory of reciprocal surfaces is connected with the theory of curves, with the theory of congruences, with the theory of complexes, and with the theory of transformations.

48. The theory of reciprocal surfaces may be applied to the determination of the singularities of a surface which is given by its equation. For instance, if the surface is of the order n, and has a nodal curve of the order a and a cuspidal curve of the order κ, then the singularities of the reciprocal surface may be determined by means of the Plückerian equations.

49. The theory may also be applied to the determination of the singularities of a surface which is the envelope of a system of planes. For instance, if the surface is the envelope of a system of planes which depend on two parameters, then the singularities of the surface may be determined by means of the theory of reciprocal surfaces.

50. The theory of reciprocal surfaces is also connected with the theory of curves. For instance, the reciprocal of a curve is a developable surface; and the singularities of the curve correspond to the singularities of the developable. The theory of reciprocal curves is therefore a particular case of the theory of reciprocal surfaces.

51. The theory may also be extended to higher dimensions. For instance, in four-dimensional space, the reciprocal of a surface is a surface; and the singularities of the two surfaces are reciprocal to each other. The theory of reciprocal surfaces in four-dimensional space is therefore analogous to the theory of reciprocal curves in three-dimensional space.

52. The Plückerian equations for a surface of the order n with a nodal curve of the order a and a cuspidal curve of the order κ may be written in the form:

$$\begin{aligned}
&q = n(n-1)^2 - a - 2\kappa,\\
&r = \tfrac{1}{2}n(n-1)(n-2)(n-3) - (n-2)a + \tfrac{1}{2}\kappa(\kappa-1),\\
&s = 3n(n-2) - 2a + 3\kappa.
\end{aligned}$$

These three equations determine the class q and the orders r and s of the nodal and cuspidal tdevelopables of the reciprocal surface, in terms of the order n of the given surface and the orders a and κ of its nodal and cuspidal curves.

53. The equations may also be written in the form:

$$\begin{aligned}
&n = q(q-1)^2 - r - 2s,\\
&a = q(q-1)(q-2) - 2r + s,\\
&\kappa = \tfrac{1}{2}s(s-1).
\end{aligned}$$

These three equations determine the order n and the singularities a and κ of the given surface, in terms of the class q and the orders r and s of the nodal and cuspidal tdevelopables of the reciprocal surface. The six equations of Nos. 52 and 53 are equivalent to each other; and they may be used to determine the singularities of a surface and its reciprocal.